

Mixed boundary characters and the integral collision extension

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Abstract

The existing all-prime boundary character and collision Frobenius define a two-term comparison over the original integral coefficient lattice. Its full tensor cokernels and torsion pairings are calculated. The order-nine cotangent extension has zero connecting map in count one and a nonzero order-three image in every count at least two. An explicit torsion character detects the image, while the retained cotangent injection has a precisely determined chain-lifting obstruction and, wherever the lift exists, kills that image. We also compute the actual Breuil–Kisin twist at the collision prism with ideal $(3 + \epsilon)$, its local divided-Frobenius domain, and the exact map from the earlier comparison complex. The former connecting class is a boundary there; different constant and T -coordinate filtration defects survive. Their exact return through local completion is also computed: the transported filtration preserves them, while reconstructing the filtration from the quotient ring loses precisely these groups. All support labels remain. These are coefficient and local prism calculations, not a computation of absolute syntomic cohomology or a Weil positivity theorem.

1 Mixed boundary characters and the collision extension

This calculation applies the existing collision differential extension to the computed characters of the all-prime mixed boundary module. The two operators retain their distinct origins: the boundary operator is an idele dilation; the collision operator is a specified Frobenius lift. We construct their exact comparison complex and its extension classes. No identification of these origins, arithmetic purity, or sign of the original Weil pairing is presumed.

1.1 The original boundary and its integral coefficient maps

Let $\mathcal{P}$ be all rational primes, $r \geq 2$, $1 \leq m < r$, and $j \geq 1$. Retain

$$\begin{aligned} \mathcal{A}_{r,m} &= \{A \subset \{1, \dots, r\} : |A| = m\}, & X_{r,m} &= \mathcal{A}_{r,m}^{\{\infty\} \sqcup \mathcal{P}}, \\ E_{r,m} &= \{(A, A, \dots) : A \in \mathcal{A}_{r,m}\}, \\ I_{r,m}(R) &= \{f \in \text{LC}(X_{r,m}, R) : f|_{E_{r,m}} = 0\}, \\ B_{r,m,j}(R) &= I_{r,m}(R) \otimes_R \Lambda^j \left(\bigoplus_p R\vartheta_p \right). \end{aligned} \tag{BCE1}$$

Locally constant means depending on finitely many place coordinates; the discrete coefficient ring is indicated. These coordinates are evaluation/integration choices at each place, not a replacement for the coefficient-support lattice L .

The original GSB8, GSB15 and GSB16 calculation gives an injection

$$\partial_{r,j}^\theta : B_{r,m,j}(\mathbb{C}) \longrightarrow H_{j-1}(\mathbb{Q}^\times, K_r^\theta), \quad U_a \partial_{r,j}^\theta(b) = \left(|a_\infty|_\infty \prod_p |a_p|_p \right)^m \partial_{r,j}^\theta(b). \tag{BCE2}$$

Here the original Schwartz source is $T = \mathcal{S}(\mathbb{A}_{\mathbb{Q}})$, $E\xi(x) = \sum_{q \in \mathbb{Q}^\times} \xi(qx)$, and $b : T \rightarrow \text{LC}(\{0, 1\}^{\{\infty\} \sqcup \mathcal{P}}, \mathbb{C})$ tensors evaluation at zero and additive integration. With algebraic tensors,

$K_r^\theta = \ker b^{\otimes r} \cap \ker E^{\otimes r}$. The source of this connecting map is the group homology of the kernel of the endpoint evaluations. The complete proof, including topology, all-prime representatives and injectivity, is GSB1–22a. Its original adelic conventions are those of Connes and Connes–Consani–Marcolli [1, 2]. BCE2 is an input from that proof, not a new assertion of integral Schwartz exactness.

Put $R_0 = \mathbb{Z}_{(3)}$. For every coefficient ring R over R_0 , there is an exact scalar-extension identification

$$I_{r,m}(R_0) \otimes_{R_0} R = I_{r,m}(R), \quad B_{r,m,j}(R_0) \otimes_{R_0} R = B_{r,m,j}(R). \quad (\text{BCE3})$$

Here both source modules over R_0 are free. To prove all these claims, order the primes and use successively the first n . At each finite stage, a basis for the kernel of endpoint restriction consists of the indicator functions of patterns other than the constant patterns. Restriction to constant patterns is exactly endpoint evaluation, so no other relation occurs. Refinement sends each old basis vector to the sum of its disjoint child patterns. Choosing one child of each old nonconstant pattern gives an explicit linear retraction. Each such block quotient is free by eliminating that one child, and new nonconstant children of old constant patterns are additional independent basis vectors. Thus each stage splits as the preceding stage plus a finite free complement. Their union is free. This proof works over every R and proves the first equality in BCE3. The exterior factor has the basis of increasing distinct prime wedges; tensoring these bases proves the second equality.

Consequently the following are actual coefficient arrows:

$$B_{r,m,j}(R_0) \longrightarrow B_{r,m,j}(\mathbb{C}) \xrightarrow{\partial_{r,j}^\theta} \text{im } \partial_{r,j}^\theta, \quad B_{r,m,j}(R_0) \longrightarrow B_{r,m,j}(\mathbb{Z}_3). \quad (\text{BCE4})$$

Both first coefficient maps are injective by the free bases just proved. There is no arrow $\mathbb{C} \rightarrow \mathbb{Q}_3$ in BCE4.

Choose the actual idele $\alpha = (3, (1)_p)$. Its modulus is 3, so its action in BCE2 is $a_m = 3^m$. This is *not* the diagonal rational idele 3, whose modulus is $3 \cdot 3^{-1} = 1$. The lattice in BCE4 is stable under U_α and its nonnegative powers. Its action is not surjective there; the inverse requires inverting 3. The entire complex idele representation in BCE2 is retained, and is not replaced by this integral semigroup lattice.

1.2 The retained collision coefficients

Write $O = \mathbb{Z}_3$, $K = \mathbb{Q}_3$, retain $c \in O$, and put

$$\begin{aligned} C &= O[\epsilon, T]/(\epsilon^2, T^2 - 6\epsilon), \quad z = \epsilon T, \quad M = O\epsilon \oplus OT \oplus Oz, \\ F_M(T) &= 9cz, \quad F_M(\epsilon) = F_M(z) = 0. \end{aligned} \quad (\text{BCE5})$$

The original algebra has basis $1, \epsilon, T, z$, by successive monic division. In particular $F_M^2 = 0$; the algebra map is the identity on O . These are the original collision coefficients of CFF1–4, not a Frobenius assigned to BCE2.

The cotangent differential has presentation

$$C^2 \xrightarrow{\begin{pmatrix} 2\epsilon & -6 \\ 0 & 2T \end{pmatrix}} C d\epsilon \oplus C dT.$$

It yields the exact O -module and extension

$$\begin{aligned} \Omega &= Ou \oplus Ov \oplus Oq \oplus (O/3)w, \\ u &= d\epsilon, \quad v = dT, \quad q = \epsilon dT, \quad w = T d\epsilon - 2\epsilon dT, \\ 0 &\longrightarrow M \xrightarrow{d_M} \Omega \xrightarrow{\rho} Q = O/9 \longrightarrow 0, \\ d_M(\epsilon) &= u, \quad d_M(T) = v, \quad d_M(z) = w + 3q, \\ \rho(u) &= \rho(v) = 0, \quad \rho(q) = 1, \quad \rho(w) = -3, \quad 9q = 3d_M(z). \end{aligned} \quad (\text{BCE6})$$

For verification directly from the eight generators of $C d\epsilon \oplus C dT$, the relation columns and their multiples impose $\epsilon u = zu = 0$, $Tv = 3u$, $zv = 0$, $3(Tu - 2\epsilon v) = 0$. All remaining multiples follow from these. Eliminating the four zero or dependent generators and setting $Tu = w + 2q$ gives exactly the displayed three free generators and the order-three generator. The image of $a\epsilon + bT + tz$ has free coordinates $a, b, 3t$; it is zero only for $a = b = t = 0$. In its cokernel, $u = v = 0$, $w = -3q$ and $3w = 0$, giving precisely $O/9$. Changing the lift q changes $3z$ by $9M$. Thus its extension class is $[3z] \in M/9M$, of exact order three.

The actual induced operators on this sequence are

$$F_\Omega(u) = F_\Omega(q) = F_\Omega(w) = 0, \quad F_\Omega(v) = 27cq, \quad F_Q = 0, \quad F_\Omega d_M = d_M F_M. \quad (\text{BCE7})$$

Indeed $d(9cz) = 9c(w + 3q) = 27cq$, since $3w = 0$. All operators square to zero. The coefficients and formulas in BCE5–7 are already defined over $\mathbb{Z}_{(3)}[c]$. They can be specialized separately to a chosen $c \in O$ and to a chosen complex parameter. Such specializations are not a map from one coefficient field to the other.

1.3 The exact operator comparison and all tensor orders

For an O -module E with endomorphism F_E , define

$$\mathcal{K}_m(E) = [E \xrightarrow{L_{m,E} = 3^m I - F_E} E] \quad \text{in cohomological degrees } 0, 1. \quad (\text{BCE8})$$

This is a specified comparison object between the two endomorphisms. Its degree-zero cohomology is the actual intertwiner module $\text{Hom}_{O[t]}(O_{3^m}, E_F)$, where t acts on the source as 3^m and on the target as F_E . Its degree-one cohomology is $\text{Ext}_{O[t]}^1(O_{3^m}, E_F)$, and the higher Ext groups are zero. Here is the proof, including signs. There is an exact free resolution

$$0 \longrightarrow O[t] \xrightarrow{3^m - t} O[t] \longrightarrow O_{3^m} \longrightarrow 0. \quad (\text{BCE9})$$

Division by the monic polynomial $t - 3^m$ proves the cokernel and the leading term proves injectivity. Applying $\text{Hom}(-, E)$ gives BCE8 with the displayed sign. More concretely, a class $x \in E/(3^m - F_E)E$ gives the extension

$$E \oplus O\tilde{1}, \quad t\tilde{1} = 3^m \tilde{1} - x, \quad t|_E = F_E.$$

Every extension has this form because its underlying O -module extension splits. Replacing $\tilde{1}$ by $\tilde{1} + y$ changes x to $x + (3^m - F_E)y$. This proves the classification and its splitting criterion without omitting the integral lattice.

For $k \geq 1$ put $M_k = M^{\otimes k}$, with $F_k = F_M^{\otimes k}$, on the original words in ϵ, T, z . Let $n = 3^k$, $a = 3^m$, and $b = (9c)^k$. The map F_k sends $T^{\otimes k}$ to $bz^{\otimes k}$ and kills every other word. Thus

$$L_{m,k} = aI - F_k, \quad L_{m,k}^{-1} = a^{-1}I + a^{-2}F_k \quad \text{over } K. \quad (\text{BCE10})$$

Multiplying the two displayed operators gives I , since $F_k^2 = 0$. In particular $\ker L_{m,k} = 0$ on the original free O -module, and BCE10 is a contracting homotopy for its rationalized two-term complex, not a deletion of its integral cokernel.

Retain $v_3(0) = +\infty$, and write $\nu = \min\{m, 2k + kv_3(c)\}$, interpreting the second term as $+\infty$ when $c = 0$. The full answer is

$$H^1 \mathcal{K}_m(M_k) \cong (O/3^m)^{3^k-2} \oplus O/3^\nu \oplus O/3^{2m-\nu}. \quad (\text{BCE11})$$

The two exceptional coordinates have the retained matrix $\begin{pmatrix} a & -b \\ 0 & a \end{pmatrix}$ in the order $z^{\otimes k}, T^{\otimes k}$. If $a \mid b$, add b/a times the first column to the second, obtaining aI . Otherwise $b = 3^\nu u$ with u a unit and $\nu < m$. Move the entry $-b$ to the first diagonal position, multiply that row by its unit inverse, and use the fact that it divides all entries to eliminate its row and column. The remaining diagonal

entry has valuation $2m - \nu$, since the original determinant is a^2 . All these changes are invertible over O . The other $3^k - 2$ coordinates are multiplication by a . This proves BCE11, including $m = 0$, $c = 0$ and all tensor orders. The row and column maps do not replace the original coordinates when using the connecting maps below.

There is also a perfect torsion pairing, with the transpose coefficient object retained:

$$\text{coker } L_{m,k} \times \text{coker } L_{m,k}^t \longrightarrow K/O, \quad ([x], [y]) \longmapsto y^t L_{m,k}^{-1} x \pmod{O}. \quad (\text{BCE12})$$

Changing x by $L_{m,k}u$ adds $y^t u \in O$; changing y by $L_{m,k}^t v$ adds $v^t x \in O$. If a class pairs to zero against all integral y , every coordinate of $L_{m,k}^{-1}x$ is integral, so x is in $L_{m,k}M_k$. The same argument with the transpose proves the other radical is zero. BCE11 gives equal finite orders on both sides, and their characters into K/O have those same orders: for a summand $O/3^h$, its character is determined by an element of $3^{-h}O/O$, also of order 3^h . Thus the pairing is perfect. It is a torsion-valued pairing, not a real ordered or Hermitian form.

1.4 Applying the order-nine extension: zero and nonzero images

By BCE7, the original exact sequence BCE6 gives a short exact sequence of the actual comparison complexes:

$$0 \longrightarrow \mathcal{K}_m(M) \xrightarrow{d_M} \mathcal{K}_m(\Omega) \xrightarrow{\rho} \mathcal{K}_m(Q) \longrightarrow 0. \quad (\text{BCE13})$$

For $m \geq 1$, the complete kernels needed for the connecting map are

$$H^0 \mathcal{K}_m(M) = 0, \quad H^0 \mathcal{K}_m(\Omega) = (O/3)w, \quad H^0 \mathcal{K}_m(Q) = Q[3^m] = 3^{\max(0, 2-m)}O/9. \quad (\text{BCE14})$$

To check the middle equality, write $xu + yv + hq + tw$. Its image has free coordinates $3^m x$, $3^m y$, $3^m h - 27cy$; all vanish only for $x = y = h = 0$. Multiplication by 3^m kills w . For $m = 0$ all three comparison maps are invertible and every cohomology group is zero.

Let $s = \max(0, 2 - m)$, and take 3^s as the displayed generator of the last module of BCE14. Its lift in Ω is $3^s q$. Its differential is

$$(3^m - F_\Omega)(3^s q) = 3^{m+s} q = 3^{m+s-1} d_M(z).$$

The sign convention BCE8 therefore gives

$$\delta_m([3^s]) = [3^{m+s-1}z] \in M/(3^m - F_M)M. \quad (\text{BCE15})$$

The class $[z]$ has exact order 3^m : if a multiple of z is $L_{m,M}(x\epsilon + yT + hz)$, its T and ϵ coefficients force $x = y = 0$, and its z coefficient is $3^m h$. Consequently

$$\delta_1 = 0, \quad \delta_m(1) = [3^{m-1}z] \neq 0, \quad 3\delta_m(1) = 0, \quad \ker \delta_m = 3O/9 \quad (m \geq 2). \quad (\text{BCE16})$$

This is independent of the retained parameter c . The kernel statement also matches the preceding map in the exact sequence: the actual map $(O/3)w \rightarrow Q[3^m]$ sends w to -3 .

For clarity the full first mixed case $m = 1$ is

$$\begin{aligned} H^1 \mathcal{K}_1(M) &= (O/3)\{\epsilon, T, z\}, & H^1 \mathcal{K}_1(\Omega) &= (O/3)\{u, v, q, w\}, \\ H^1 \mathcal{K}_1(Q) &= O/3, & d_{M,*}(\epsilon) &= u, \quad d_{M,*}(T) = v, \quad d_{M,*}(z) = w, \\ & & \rho_* q &= 1, \quad \rho_* u = \rho_* v = \rho_* w = 0. \end{aligned} \quad (\text{BCE17})$$

The free image in Ω is exactly $3Ou + 3Ov + 3Oq$: the off-diagonal $27cq$ lies in it, and no w lies in the image. This proves BCE17 and its zero connecting map directly, rather than by forgetting the original order-nine quotient. For $m \geq 1$ in general the same argument and two-by-two elimination give

$$\begin{aligned} H^1 \mathcal{K}_m(\Omega) &\cong O/3^m \oplus O/3^\mu \oplus O/3^{2m-\mu} \oplus O/3, \\ \mu &= \min\{m, 3 + v_3(c)\}, & H^1 \mathcal{K}_m(Q) &= O/3^{\min(m, 2)}. \end{aligned} \quad (\text{BCE18})$$

The original map is still BCE6; BCE18 does not authorize replacing it by an unspecified map between cyclic presentations.

The z -coordinate functional in BCE12 is especially explicit:

$$\lambda_m([x_\epsilon\epsilon + x_T T + x_z z]) = \frac{x_z}{3^m} + \frac{9c x_T}{3^{2m}} \pmod{O}. \quad (\text{BCE19})$$

Its value on $L_{m,M}y$ is $y_z \in O$, so it is well-defined. It detects the whole displayed connecting generator:

$$\lambda_m(\delta_m(1)) = \frac{1}{3} \pmod{O} \neq 0 \quad (m \geq 2). \quad (\text{BCE20})$$

Thus rational acyclicity in BCE10 is compatible with an integral extension and its explicit nonzero torsion character. An additive map from this order-three image to a characteristic-zero vector space is zero, since its target has no element killed by 3. This assertion concerns that specified additive map, not all possible arithmetic uses of the extension.

1.5 The existing cotangent injection and its exact repair

Retain $H = Ov_1 \oplus Ov_2 \oplus Ov_3 = H^{-1}(L_{C/O})$. The original cotangent cycles are

$$v_1 = (\epsilon, 0), \quad v_2 = (z, 0), \quad v_3 = (3T, z).$$

They exhaust the kernel of the matrix preceding BCE6: its lower equation forces its second coordinate to be a multiple of z , and its upper equation then fixes the T -coefficient of its first coordinate. The actual Frobenius on H is zero. Indeed the polynomial lift sends its two relation generators $\epsilon^2, T^2 - 6\epsilon$ to $0, 81c^2T^2\epsilon^2$. The induced degree-minus-one semilinear matrix is $\begin{pmatrix} 0 & 486c^2\epsilon \\ 0 & 0 \end{pmatrix}$; the second coordinates of the displayed cycles are $0, 0, z$, whose Frobenius values all vanish.

The retained injection and defect are

$$\begin{aligned} \kappa(\epsilon) &= 3v_1, & \kappa(T) &= v_3 - 2v_2, & \kappa(z) &= 3v_2, & F_H &= 0, \\ \Delta &= \kappa F_M - F_H \kappa, & \Delta(T) &= 27cv_2, & \Delta(\epsilon) &= \Delta(z) = 0. \end{aligned} \quad (\text{BCE21})$$

The free coefficients prove injectivity and the displayed defect follows by applying F_M to T .

There is a chain map $\mathcal{K}_m(M) \rightarrow \mathcal{K}_m(H)$ with its degree-zero component exactly the original κ if and only if

$$27c \in 3^m O. \quad (\text{BCE22})$$

Its degree-one component, when it exists, is unique and is

$$\kappa_m^1 = \kappa + 3^{-m} \kappa F_M, \quad \kappa_m^1(T) = v_3 - 2v_2 + 3^{3-m} cv_2, \quad \kappa_m^1(\epsilon) = 3v_1, \quad \kappa_m^1(z) = 3v_2. \quad (\text{BCE23})$$

The chain equation is $\kappa_m^1(3^m - F_M) = 3^m \kappa$. Over K , BCE10 forces BCE23, including its positive correction. It is integral exactly under BCE22, and multiplication proves the equation directly. Because the source and target are free, the rational uniqueness also proves integral uniqueness. For $c \neq 0$, BCE22 is $m \leq 3 + v_3(c)$; for $c = 0$ it holds for every m .

At larger m , the exact obstruction to this fixed-degree-zero lift is

$$\text{ob}_m(\kappa) : M \longrightarrow H/3^m H, \quad T \longmapsto 27cv_2 \pmod{3^m H}, \quad \epsilon, z \longmapsto 0. \quad (\text{BCE24})$$

For $c \neq 0$ and $m > 3 + v_3(c)$, its image has exact order $3^{m-3-v_3(c)}$. This follows by the valuation of its one nonzero coefficient in the free v_2 coordinate. It is the obstruction to the specified lift, not to every possible comparison between these objects.

Where BCE23 exists and $m \geq 2$, its induced degree-one map has the evaluated composition

$$(\kappa_m^1)_* \delta_m(1) = [3^{m-1} \kappa_m^1(z)] = [3^m v_2] = 0 \quad \text{in } H/3^m H. \quad (\text{BCE25})$$

The nonzero class detected in BCE20 is therefore lost by this particular cotangent receiver. The nonzero residue character and the zero receiver are both retained. The chain lift has two different components; it does not remove the original coefficient Frobenius defect. Indeed $\kappa_m^1 F_M(T) = 27cv_2$ still, while $F_H = 0$.

1.6 All-prime boundary families and the full support space

Let $B = B_{r,m,j}(O)$ with $t = 3^m$. By BCE3 it is free. The analogue of BCE9 with $O[t] \otimes_O B$ gives

$$\begin{aligned} \mathrm{RHom}_{O[t]}(B, E_F) &= \left[\mathrm{Hom}_O(B, E) \xrightarrow{3^m - F_E} \mathrm{Hom}_O(B, E) \right], \\ \mathrm{Ext}_{O[t]}^1(B, M_k) &= \mathrm{Hom}_O(B, \mathrm{coker} L_{m,k}), \quad \mathrm{Hom}_{O[t]}(B, M_k) = 0. \end{aligned} \quad (\text{BCE26})$$

Injectivity of the resolution is checked by the highest polynomial coefficient. Projectivity of B implies that $\mathrm{Hom}_O(B, -)$ is exact, proving the cokernel identity. For an infinite basis this Hom is a product, not a tensor direct sum; the formula retains that distinction. Applying this exact functor to BCE13–25 gives their entire all-prime family, not a finite set of tested characters.

In particular choose distinct $A, D \in \mathcal{A}_{r,m}$, a prime p , and the balanced point with infinity coordinate A , p -coordinate D , and every other coordinate A . For $j = 1$, define $\ell : B_{r,m,1}(O) \rightarrow O$ by taking the ϑ_p -coefficient and evaluating it at this point. The cylinder $f = \mathbf{1}_{\{x_\infty=A, x_p=D\}}$ vanishes on every endpoint and satisfies $\ell(f \otimes \vartheta_p) = 1$. For $m \geq 2$, the input $\ell \bmod 9 \in \mathrm{Hom}_O(B, Q)$ therefore has connecting class

$$b \mapsto [3^{m-1} z \ell(b)], \quad \lambda_m \delta_m(\ell)(f \otimes \vartheta_p) = 1/3 \pmod{O}. \quad (\text{BCE27})$$

For $m = 1$ every connecting class is zero, by BCE16. The first nonzero count is $m = 2$, first present at $r = 3$; one may take $A = \{1, 2\}$, $D = \{1, 3\}$. Each rational prime supplies such a witness. This describes the actual integral character module embedded by BCE4; it does not claim a new Frobenius on the original Schwartz source.

The relation to the full support spectrum is also explicit. For any finite nontrivial bounded distributive L , let $Y = \mathrm{Spec} G_L(\mathbb{Z}) = S \triangleright \mathrm{Spec} \mathbb{Z}$, as in FL1–15 and CFF28–35. The exact constant-sheaf coupling $\mathcal{U}(E) = (c_X E, c_S E, \Delta)$ places the original coefficient module on every stalk. Its ordinary global cohomology is $E[0]$; its supported cochain complex is $P_S \otimes_O E$, where

$$P_S = \mathrm{Cone}(O[0] \xrightarrow{\Delta} C^\bullet(S, O))[-1], \quad P_S^0 = O, \quad P_S^i = C^{i-1}(S, O) \ (i \geq 1),$$

with differential $-\Delta$ followed by $-d$. These are the exact proved CFF31 maps; all terms of P_S are finite free. Hence applying $\mathcal{U}$, or $P_S \otimes_O -$, to BCE13 gives respectively the stalkwise extension or the short exact sequence of supported total complexes. The total differential is $d_P \otimes 1 + (-1)^i 1 \otimes d_K$ on bidegree $(i, *)$; thus the two cone signs are fixed. No nonvanishing in supported cohomology is inferred merely from BCE27. For the two-element lattice, $P_S = [O \xrightarrow{-1} O]$ is contractible, and its tensor complex is contractible as well: $h_P \otimes 1$ contracts the displayed total differential, since the mixed terms cancel with their opposite signs. Ordinary cohomology and all stalks still retain BCE13–27.

Finally the separate labelled carrier is

$$G_L(E) = \{(v, 1_L) : v \in E\} \cup \{(0, \lambda) : \lambda \in L\}, \quad G_L(f)(v, \lambda) = (fv, \lambda). \quad (\text{BCE28})$$

Addition uses the original join of labels. Evaluating the two types of element proves that every additive map above lifts, preserves composition, and fixes all zero labels. A nonzero class sent to amplitude zero has image $e = (0, 1_L)$, while $\tau = (0, 0_L)$ remains distinct. The torsion detector lifts to $G_L(K/O)$ with the same labels. No ordering or positivity on this torsion target has been invented.

1.7 The actual prism ideal and its Breuil–Kisin twist

The comparison BCE8 is not the definition of syntomic cohomology. Bhatt–Lurie’s construction uses the absolute Nygaard filtration, the Breuil–Kisin twist, and the difference between its partially defined Frobenius and inclusion [4]. We now calculate the corresponding *local twisted fixed-point*

complex at the given prism. This terminology specifies the calculation's domain: we do not identify sections at one prism with the derived global sections of the absolute prismatic site.

The bounded prism proved in DP1–16 is

$$(C, I), \quad I = (d), \quad d = 3 + \epsilon, \quad d(3 - \epsilon) = 9, \quad \phi(d) = 3. \quad (\text{BCE29})$$

In particular d is a nonzerodivisor. The quotient map $q : C \rightarrow O$, $q(\epsilon) = q(T) = 0$, is a map of prisms to $(O, (3))$: it commutes with Frobenius, hence with the delta-operation $(\phi(x) - x^3)/3$, and sends I to (3) . The nilpotent ideal M has $M^4 = 0$, and $\phi^2|_M = 0$. Neither equality identifies d with 3.

Write $L = C\{1\}$ for the Breuil–Kisin line. We use the following exact source facts: it is invertible and commutes with base change of prisms; its semilinear Frobenius linearizes to an isomorphism $\phi^*L \simeq d^{-1}L$; and the crystalline line $O\{1\}$ has its canonical generator e_O with $\phi(e_O) = e_O/3$. These are the statements labelled **remark:BK-functoriality**, **remark:Frobenius-on-twist**, and **BKcrystalline** in the original Bhatt–Lurie source.

Choose a lift $e \in L$ of e_O . It generates L : C is local with nilpotent ideal M and quotient O , and a lift of a generator modulo this ideal is a generator. Thus, for an actual unit $u \in C^\times$, $\phi(e) = ue/d$. Reduction to O gives $q(u) = 1$. Retain this initial generator, its unit, and the following explicit comparison:

$$v = u\phi(u), \quad e_* = ve, \quad \phi(e_*) = e_*/d, \quad e = v^{-1}e_*, \quad v^{-1} = \sum_{i=0}^3 (-1)^i (v-1)^i. \quad (\text{BCE30})$$

Indeed $\phi^2(u) = 1$, so $\phi(v) = \phi(u)$ and $\phi(v)u = v$. The inverse formula follows from $M^4 = 0$. This comparison is independent of the chosen lift e . If $e' = we$ with $q(w) = 1$, its unit is $u' = \phi(w)u/w$, and

$$u'\phi(u')w = u\phi(u)\phi^2(w) = u\phi(u).$$

It is also the unique generator reducing to e_O and satisfying the displayed Frobenius equation: the ratio of two such generators satisfies $\phi(w) = w$, hence $w = \phi^2(w) = q(w) = 1$. Thus BCE30 is a proved coordinate map retaining the original unit, not an omission of it or of the distinguished element.

For $m \geq 0$, use $e_*^{\otimes m}$ to display the line $C\{m\}$. Its Frobenius is exactly $x \mapsto \phi(x)/d^m$. Define

$$N_m C = \{x \in C : \phi(x) \in d^m C\}, \quad J_m(C) = [N_m C \xrightarrow{\phi/d^m - \iota} C] \quad (\text{BCE31})$$

as a complex of O -modules in degrees zero and one. The Frobenius difference is O -linear, not asserted to be C -linear. Division means the unique preimage under multiplication by the nonzerodivisor d^m . The sign is that of Bhatt–Lurie's syntomic fiber, $\phi\{m\} - \iota$. This defines a local complex without assuming a descent theorem for it.

1.8 Complete local filtration and its quotient defect

For $m \geq 1$, let

$$\eta_m = \begin{cases} 1, & 3 \nmid m, \\ 0, & 3 \mid m, \end{cases} \quad h_m = \max\{0, m - 2 - v_3(c)\},$$

where $h_m = 0$ if $c = 0$. In the original ordered coordinates $1, \epsilon, T, z$ the full result is

$$N_m C = 3^{m+\eta_m} O 1 \oplus O\epsilon \oplus 3^{h_m} OT \oplus Oz, \quad N_m M = O\epsilon \oplus 3^{h_m} OT \oplus Oz. \quad (\text{BCE32})$$

Here $N_m M = N_m C \cap M$, and its divided Frobenius has image in M . To prove every coefficient, retain

$$d^m = 3^m + m3^{m-1}\epsilon. \quad (\text{BCE33})$$

This is the full binomial expansion because $\epsilon^2 = 0$. More explicitly, for $x = x_0 + x_\epsilon \epsilon + x_T T + x_z z$ one has

$$\phi(x) = x_0 + 9cx_T z, \quad d^{-m}\phi(x) = 3^{-m}x_0 - m3^{-m-1}x_0\epsilon + 3^{2-m}cx_T z \quad \text{in } C[1/3].$$

All these coordinates are integral precisely when

$$x_0 \in 3^m O, \quad mx_0 \in 3^{m+1} O, \quad 9cx_T \in 3^m O.$$

The first two conditions are $x_0 \in 3^{m+\eta_m} O$; the third is $x_T \in 3^{h_m} O$. No conditions are imposed on x_ϵ, x_z , proving BCE32.

Write a general element of the domain in its retained basis as $3^{m+\eta_m}a + b\epsilon + 3^{h_m}tT + fz$. The complete differential is

$$\begin{aligned} (\phi/d^m - \iota)(x) &= 3^{\eta_m}(1 - 3^m)a \\ &\quad + (-m3^{\eta_m-1}a - b)\epsilon - 3^{h_m}tT + (3^{2+h_m-m}ct - f)z. \end{aligned} \quad (\text{BCE34})$$

The apparent negative powers are integral by the definitions of η_m, h_m . In particular $m/3 \in O$ when $\eta_m = 0$. The factor $1 - 3^m$ is a unit. The constant coordinate in BCE34 first forces $a = 0$ for an element of its kernel; then the T coordinate forces $t = 0$, and the other coordinates force $b = f = 0$. The independent columns for ϵ, z kill those coordinates in the cokernel. The two remaining relations are exactly $3^{\eta_m}(1 - 3^m)$ in the original constant coordinate and -3^{h_m} in the original T coordinate. Consequently

$$\begin{aligned} H^0 J_m(C) &= H^0 J_m(M) = 0, \\ H^1 J_m(C) &= (O/3^{\eta_m})[1] \oplus (O/3^{h_m})[T], \quad H^1 J_m(M) = (O/3^{h_m})[T]. \end{aligned} \quad (\text{BCE35})$$

These identifications are the coordinate maps $[x] \mapsto (x_0 \bmod 3^{\eta_m}, x_T \bmod 3^{h_m})$; the original off-diagonal terms remain in BCE34. For $m = 0$, $N_0 C = C$ and $\phi - I$ is zero on O and invertible on M , with inverse $-I - \phi$. Thus $H^0 J_0(C) = H^1 J_0(C) = O$, while $J_0(M)$ is contractible.

The constant-coordinate torsion in BCE35 has an exact geometric source: the filtration does not surject under the original prism quotient q . Indeed

$$N_m O = 3^m O, \quad q(N_m C) = 3^{m+\eta_m} O, \quad N_m O / q(N_m C) = 3^m O / 3^{m+\eta_m} O. \quad (\text{BCE36})$$

The quotient complex $J_m(C)/J_m(M)$ is

$$\bar{J}_m = [3^{m+\eta_m} O \xrightarrow{3^{-m}-1} O].$$

There is an exact sequence of complexes

$$0 \longrightarrow \bar{J}_m \longrightarrow J_m(O) \longrightarrow [3^m O / 3^{m+\eta_m} O \longrightarrow 0] \longrightarrow 0. \quad (\text{BCE37})$$

Here $J_m(O)$ is contractible since its differential, on the generator 3^m , has value $1 - 3^m$, a unit. The connecting map takes the class of 3^m to $[1 - 3^m] \in H^1 \bar{J}_m$, an isomorphism. Thus the order-three constant class, present exactly when $3 \nmid m$, is the calculated defect of filtered surjectivity. It would be lost by replacing $3 + \epsilon$ with 3 before doing the calculation.

1.9 The exact return from the earlier collision comparison

There is a chain map retaining both complexes and signs:

$$\begin{array}{ccc} M & \xrightarrow{3^m - \phi} & M \\ 3^m \downarrow & & \downarrow -1 \\ N_m M & \xrightarrow{\phi/d^m - \iota} & M. \end{array} \quad (\text{BCE38})$$

For its domain, $\phi(M) \subset Oz$ and $d^m z = 3^m z$. Hence $3^m M \subset N_m M$, and the lower differential on $3^m x$ is $\phi(x) - 3^m x$, which proves the square. For $m \geq 1$, the map on degree-one cohomology is the explicit surjection

$$M/(3^m - \phi)M \longrightarrow O/3^{h_m}, \quad [x] \longmapsto -x_T \bmod 3^{h_m}. \quad (\text{BCE39})$$

Well-definedness also follows directly from $h_m \leq m$ and the fact that $\phi(x)$ has no T coordinate. In particular the previous nonzero order-three connecting class has zero image, with an actual primitive:

$$-3^{m-1}z = (\phi/d^m - \iota)(3^{m-1}z), \quad m \geq 2. \quad (\text{BCE40})$$

The surviving T class when $m > 2 + v_3(c)$, and the constant class when $3 \nmid m$, have just been computed; neither is substituted for the vanished z class. Applying $\text{Hom}_O(B_{r,m,j}(O), -)$ preserves this exact calculation since BCE3 proves that the source is free. The chain map therefore kills the entire family BCE27, not merely one chosen prime witness. Applying the constant-sheaf coupling or the supported total-complex construction following BCE27 preserves the same chain map and primitive. The separate label functor BCE28 keeps every zero-support label on that map.

Finally we specify why BCE35 is not a computation of absolute syntomic cohomology. The original prism quotient is

$$D = C/(3 + \epsilon) \cong (O/9)[T]/(T^2), \quad D/3D = \mathbb{F}_3[T]/(T^2). \quad (\text{BCE41})$$

Frobenius on the second ring has image only the constants, so is not surjective. A quotient of a perfectoid ring has surjective Frobenius modulo 3, since roots lift from the perfectoid ring. Thus D is not semiperfectoid, and the quasiregular semiperfectoid formula for absolute prismatic cohomology cannot be applied to identify it with the given ring C . The actual absolute construction in [4], labelled `construction: absolute-Nygaard-untwisted`, retains derived global sections and a de Rham pullback; its syntomic fiber is labelled `construction: syntomic-complex`. Those terms have not been evaluated here. BCE30–40 instead compute the local twist, domain, differential, and receiving map completely. They show exactly which previous class is lost before any proposed global arithmetic comparison.

1.10 Completion retains the defect through its transported filtration

The previous computation permits an exact completion comparison, without identifying the result with absolute Nygaard completion. Put $R = \ker(\phi : C \rightarrow C)$. Directly from BCE5,

$$\begin{aligned} R &= O\epsilon \oplus Oz \quad (c \neq 0), & R &= M \quad (c = 0), \\ \widehat{C}_N &:= \varprojlim_m C/N_m C = C/R = \begin{cases} O[T]/(T^2), & c \neq 0, \\ O, & c = 0. \end{cases} \end{aligned} \quad (\text{BCE42})$$

Here equality with the quotient is the isomorphism induced by the original quotient map $q_N : C \rightarrow C/R$. Indeed BCE32 gives $\bigcap_m N_m C = R$. On C/R , the remaining coordinate exponents $m + \eta_m$ and, when present, h_m , tend to infinity and differ from m by bounded constants. Their filtration is cofinal with the 3-adic filtration of the finite free O -module C/R , which is complete. This proves both kernel and surjectivity of the inverse-limit map. The quotient tower has surjective transition maps, so its first derived inverse limit is zero as well. The original multiplication descends because R is an ideal; $T^2 = 6\epsilon$ becomes $T^2 = 0$, with its quotient map recorded.

Write $\bar{C} = C/R$. The transported filtration and induced Frobenius are exactly

$$\begin{aligned} \bar{N}_m &= q_N(N_m C) = 3^{m+\eta_m} O 1 \oplus 3^{h_m} OT, & \bar{\phi}(x_0 + x_T T) &= x_0, & q_N(d) &= 3, \\ \tilde{N}_m &:= \{x \in \bar{C} : \bar{\phi}(x) \in 3^m \bar{C}\} & &= 3^m O 1 \oplus OT. \end{aligned} \quad (\text{BCE43})$$

Every T -summand in BCE43 is omitted when $c = 0$. Thus recomputing a Frobenius preimage after taking this quotient is a different operation from transporting the original filtration. Their

domains agree exactly when $\eta_m = h_m = 0$. For $c \neq 0$, this means $3 \mid m$ and $m \leq 2 + v_3(c)$; for $c = 0$, it means $3 \mid m$. In every other case the inclusion $\bar{N}_m \subset \tilde{N}_m$ is strict.

Define $\bar{J}_m = [\bar{N}_m \xrightarrow{\bar{\phi}/3^{m-\iota}} \bar{C}]$ and $\tilde{J}_m = [\tilde{N}_m \xrightarrow{\tilde{\phi}/3^{m-\iota}} \bar{C}]$. These remain complexes of O -modules; no quotient $\bar{C}$ -module structure on their cohomology is presumed. The divided maps descend from $J_m(C)$: $\phi(R) = 0$, and applying q_N to $d^m y = \phi(x)$ gives $3^m q_N(y) = \bar{\phi}(q_N(x))$. Both quotient source and quotient target are torsionfree, so this specifies the divided map uniquely. The kernel of $J_m(C) \rightarrow \bar{J}_m$ is exactly $[R \xrightarrow{-1} R]$, proving that the quotient is a quasi-isomorphism.

For $m \geq 1$, the other complex $\tilde{J}_m$ is contractible: its differential sends 3^m to the unit $1 - 3^m$, and T , when present, to $-T$. The precise difference between the two complexes is

$$\begin{aligned} Q_m &= \tilde{N}_m / \bar{N}_m = (O/3^{\eta_m})[3^m] \oplus (O/3^{h_m})[T], \\ 0 &\longrightarrow \bar{J}_m \longrightarrow \tilde{J}_m \longrightarrow [Q_m \longrightarrow 0] \longrightarrow 0, \\ Q_m &\xrightarrow{\sim} H^1 \bar{J}_m, \quad [3^m] \longmapsto [1 - 3^m], \quad [T] \longmapsto [-T]. \end{aligned} \tag{BCE44}$$

Again omit T when $c = 0$. The exact sequence is termwise exact by its definitions. Lifting each displayed degree-zero quotient generator and applying its original differential gives the connecting formula and its signs. Contractibility of $\tilde{J}_m$ proves the isomorphism. This recovers every group in BCE35, including both summands, as the explicit defect of replacing the transported filtration. Although ϵ vanishes under completion, its η_m -contribution remains in that filtration. Thus the completed ring and endomorphism alone do not retain all the local comparison data.

There is also an explicit return and chain homotopy for the completion map $\pi : J_m(C) \rightarrow \bar{J}_m$, for $m \geq 1$. Let $s^1 : \bar{C} \rightarrow C$ lift the original coordinates $1, T$ identically, omitting T when $c = 0$. For a domain element $\bar{x} = 3^{m+\eta_m}a + 3^{h_m}tT$, put

$$\begin{aligned} s^0(\bar{x}) &= 3^{m+\eta_m}a + 3^{h_m}tT - m3^{\eta_m-1}a\epsilon + 3^{2+h_m-m}ctz, \\ P_R(y) &= y - s^1\pi(y), \quad h(y) = -P_R(y) : J_m(C)^1 \longrightarrow J_m(C)^0, \\ \pi s &= I, \quad d_J h + h d_J = I - s\pi. \end{aligned} \tag{BCE45}$$

When $c = 0$, omit both terms containing t . The added terms of s^0 lie in $R \subset N_m C$. Substituting into BCE34 cancels its two off-diagonal entries, proving $d_J s^0 = s^1 d_{\bar{J}}$. In degree one, $d_J h(y) = P_R(y)$, since $d_J|_R = -I$. In degree zero, writing x in the basis of BCE34 gives

$$-P_R(d_J x) = x - s^0\pi(x),$$

with the same two displayed corrections; for $c = 0$ the T -coordinate lies in R and obeys the same identity. These calculations prove the two homotopy identities on every original coordinate. The return is O -linear, not a section of rings. It preserves the local cohomology without pretending that the completed coefficient ring retains the original multiplication or the untransported filtration.

1.11 Exact conclusions and the remaining arithmetic comparison

The computation closes the coefficient-extension calculation for the actual balanced characters: the first mixed count kills the connecting map, the second retains an order-three class, and BCE19 detects it exactly. The old cotangent injection either has the unique integral chain repair BCE23, which kills this class, or has the evaluated obstruction BCE24. All tensor comparison groups and their perfect torsion pairings are given in BCE11–12.

These results do not impose positivity on the original zeta pairing. Their characteristic-zero comparison complexes are contractible by BCE10, while the original complex boundary module of BCE2 itself remains nonzero. Those are different objects joined by the explicit coefficient maps BCE3–4, not a license to discard the boundary. On the universal $\mathbb{Z}_{(3)}[c]$ presentation, extension to $\mathbb{C}[c]$ gives the same inverse BCE10 since 3 is invertible; extension to O retains the computed torsion. The classical zeta function, its poles, Gamma factors, completion multiplier and original

Weil form have not been replaced or altered by any of these coefficient operations. An arithmetic sign comparison must act on that form and account for these exact zero and nonzero images. No such sign conclusion follows from this calculation alone. Moreover, using the actual prism ideal and its twist changes the receiving complex as in BCE31–40: the earlier order-three class has an explicit primitive there, while different filtration defects remain. An absolute or arithmetic interpretation must retain that calculated distinction.

References

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- [4] Bhargav Bhatt and Jacob Lurie. *Absolute Prismatic Cohomology*. Original arXiv edition. Original author source `apc.tex`: the Breuil–Kisin functor, its Frobenius, crystalline generator, absolute Nygaard construction and syntomic fiber are identified by their source labels in BCE29–45. The local comparison is derived here without substituting it for absolute descent.
- [5] Programme proof, *The distinguished collision prism and its actual module specializations*, DP1–43. Complete retained proof. The source establishes the bounded prism $(C, (3 + \epsilon))$, its original Frobenius and all axioms, before the present local twisted calculation.
- [6] Programme proof, *Collision Frobenius, integral monodromy filtrations, and the full supported cohomology receiver*, CFF1–35 with CF1–67. The complete retained source `COLLISION_FROBENIUS_FILTRATION_AND_SUPPORT.tex` accompanies this source edition. Coefficient formulas needed here are proved again in BCE5–7 and BCE21.
- [7] Programme proof, *Evaluation of the entire secondary mixed-boundary map*, GSB1–22a. Complete pinned proof source.

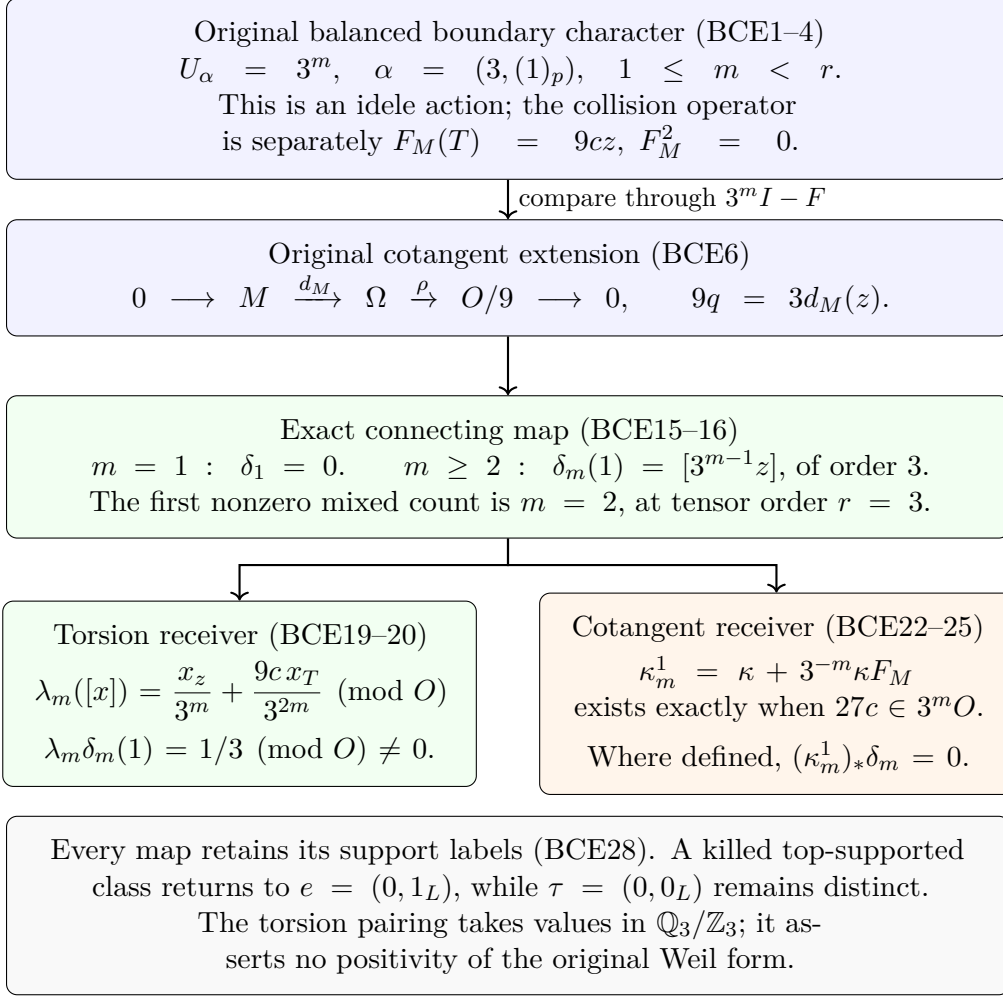


Figure 1: The two computed receivers of the mixed collision class, with original constants and signs. The top box uses the Connes–Consani–Marcolli adelic conventions through the published GSB proof; the coefficient extension is the original CFF construction. The figure depicts operator comparisons, not an identification of idele dilation with prismatic Frobenius. All proofs and exact source maps are BCE1–28.

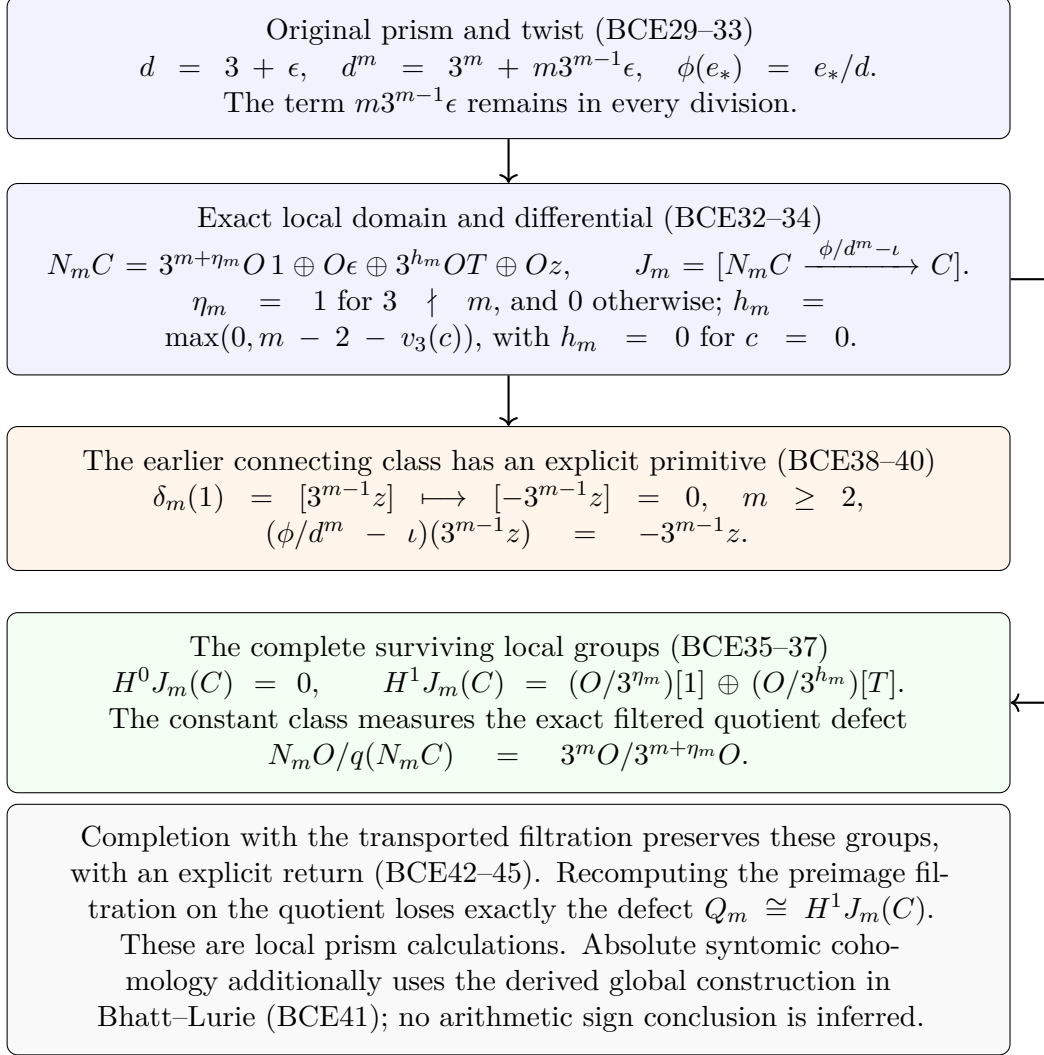


Figure 2: The local twisted receiving calculation preserves the distinguished element and changes the answer. The old z class dies by a displayed primitive; the surviving 1 and T classes have different definitions and exact origins. The diagram applies to every $m \geq 1$ and every original parameter c , not a sampled range. Full proofs and the explicit twist coordinate map are BCE29–45.